Safety Security Control Unit - IP Card

Structured design summary

Field	Value
Basic Info	
Licensing	Open-source (CERN-OHL-S-2.0)
Target FPGA or ASIC	FPGA/ASIC
Design Language	SystemVerilog
System Level Features	
Coupling Type	Tightly-coupled
Data Types	-
ISA	RV32I + Custom CSRs
Peak Ops Per Cycle	-
Interfaces	-
Supported Operations	-
Compiler Support	-
External Memory Required	-
Cache Or Scratchpad	-
Architecture	
Type	null
Number of Clock Domains	1
Control Interface > ISA Extension	Custom CSRs required
Control Interface > Memory Mapped	No
Control Interface > Protocol	CSR instructions
Memory Interface > Interface Protocol	-
Memory Interface > Hierarchy Level	-
Memory Interface > Cached	No
Memory Interface > IOMMU	No
Interrupts	Triggers Interrupts

Field	Value
Microarchitecture	
Parametrization > Parametric Number of Units	-
Parametrization > Parametric Config	RTL from a parametrized generator
Programmability > Programmable Cores	-
Programmability > ISA Extension	-
Software	
Compiler > Requires Specialized Compiler	-
Compiler > Supported Compilers	-
Hardware Abstraction Layer > Hal Functions	-
High Level API > Has High Level API	No
High Level API > SDK Repository	None
High Level API > Has Domain Specific Compiler	No
Integration	
IP Distribution > Manifest Type	Bender.yml
IP Distribution > Standalone Simulation	Yes
IP Distribution > Standalone Simulation Software Requirements	No
IP Distribution > Integration Docs Or Examples	Yes
IP Distribution > IP-XACT	-
IP Distribution > System RDL	-
Synthesis > Is Synthesizable	Yes
Synthesis > FPGA Synthesis Examples	No
Synthesis > ASIC Synthesis Examples	No
Simulation > Closed Source Simulation	Yes
Simulation > Open Source Simulation	Yes
Evaluation > PPA Results Available	No

Field	Value
Physical Implementation	
ASIC Area > Number of Gates	-
ASIC Area > On-Chip Internal Memory Size	-
ASIC Area > Target Clock Frequency	-